The Importance of Being Timely

Anticipatory cash transfers in climate disaster response

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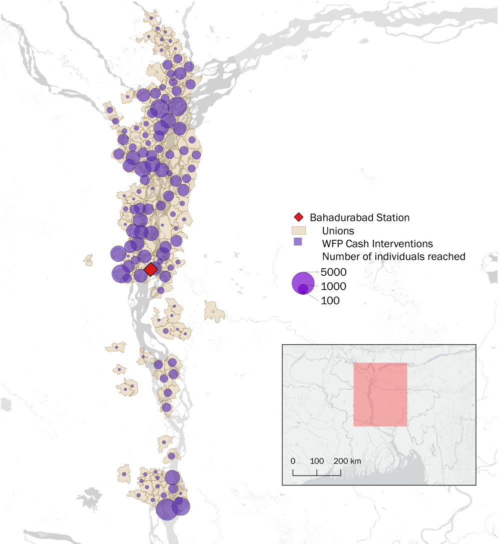
Climate change is making extreme weather events more frequent: how to improve the effectiveness of response and related financing?

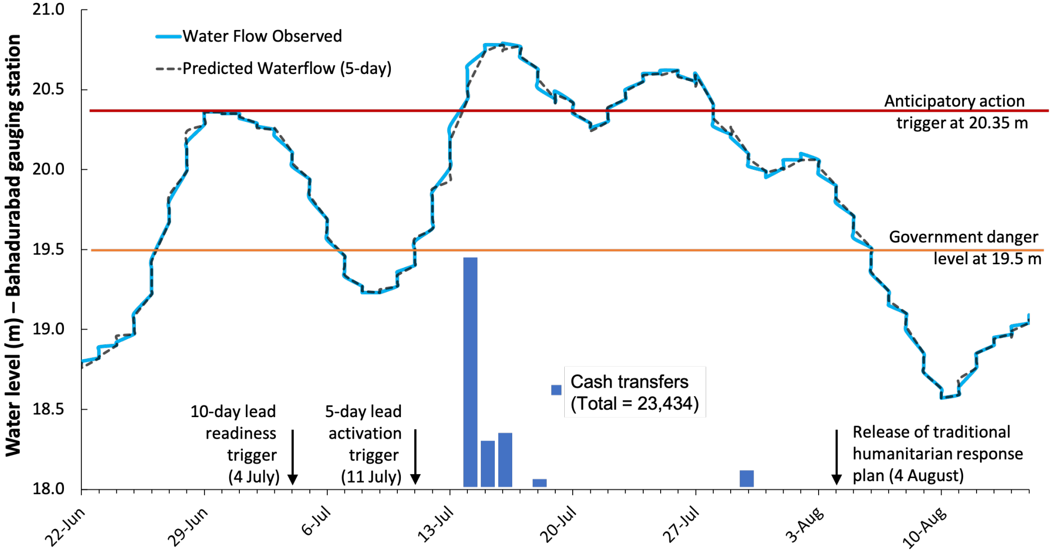
1. Anticipatory cash transfers for flood-response in Bangladesh

In 2020, Bangladesh experienced the second highest and protracted flood in the past two decades, with 5.5 million people affected



WFP sent 4,500 Taka ($53) using mobile money to 23,000 households in 131 unions along the Jamuna River in Bangladesh prior to and during the flood.

Forecast-based triggers, pre-arranged financing and five days of cash transfers



There are large evidence gaps on crisis response

- Strong evidence that *regular* cash transfers help cushion the negative effects of shocks (e.g. Janvry et al., 2006; Aker et al., 2016; Jensen et al., 2017; Adhvaryu et al., 2022; Bottan et al., 2021

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- But evidence gaps on:
  1. The impact of cash transfers delivered specifically in response to fast-onset climate hazards, especially one-off cash transfers (De Mel et al., 2012; Ivaschenko et al., 2020)
  2. Timing of cash transfers relative to climate shocks (Jeong and Trako, 2022)
  3. Impact of humanitarian interventions (Puri et al., 2017; Jeong & Trako, 202)
- We examine the impacts of a one-off cash transfer delivered to vulnerable households *forecast* to experience extreme flooding in Bangladesh.

A quasi-experimental design with three data sources

9,000 phone surveys 3 months later

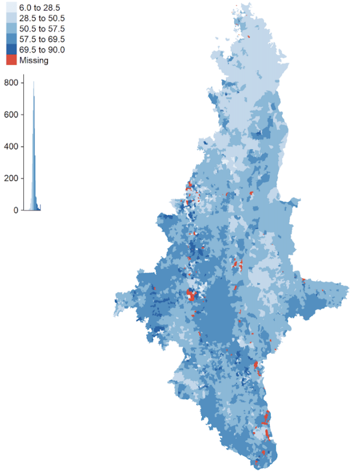
- Vulnerable households were identified in flood-affected areas using pre-existing lists, but only one mobile money provider coud be used.
- Otherwise comparable households with incorrect mobile wallets or inactive bKash accounts were excluded.

Second round of phone surveys 5 months later

Satellite flood data

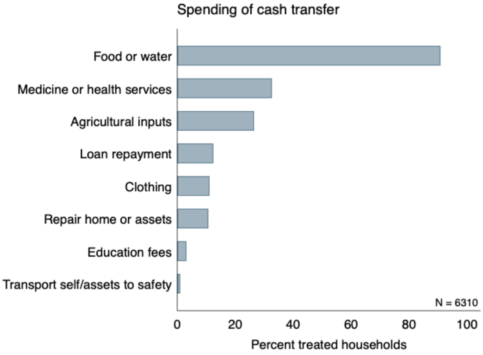
- Identifies the date of local peak flooding to complement timing analysis

Variation in local flood peak date (days since 6 June, 2020)

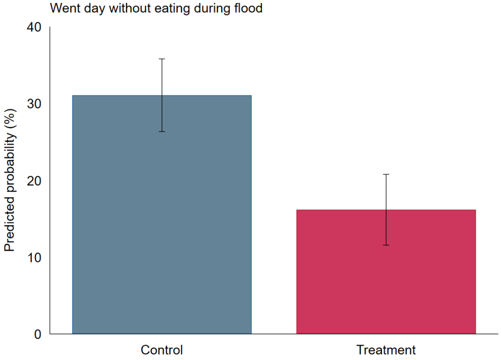


Cash was spent on food and reduced food insecurity during the floods

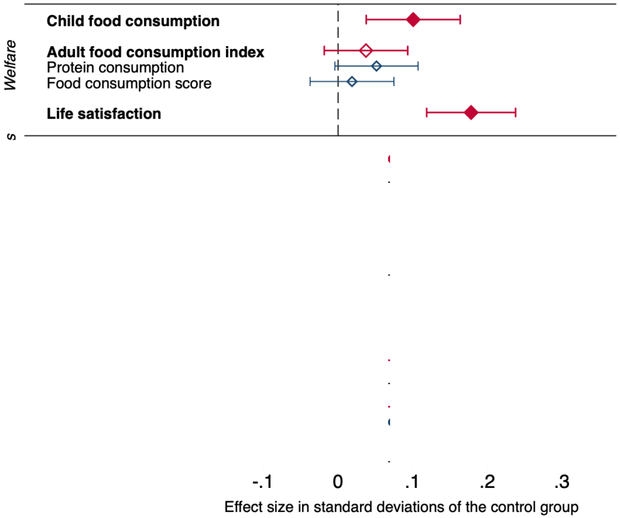
Most households used the cash transfer to buy food or water



Treated households were less likely to go without eating during the flood



Anticipatory cash improves welfare, even three months later



Control mean	Treatment effect	q-value
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39.5	+0.7%	
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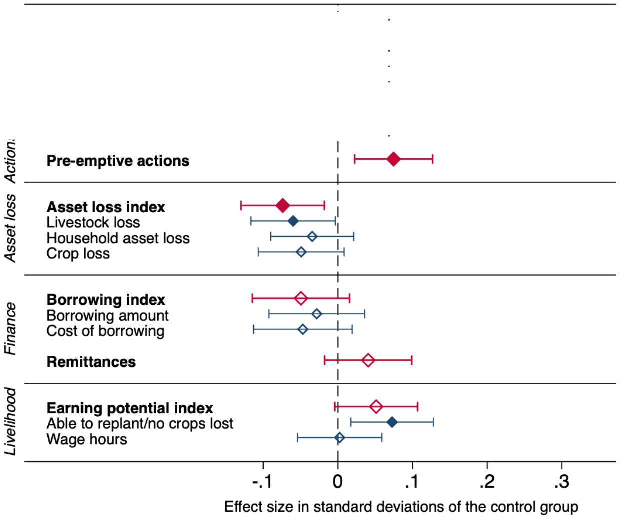

Control mean	Treatment effect	q-value
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1.31	-3.1%	
15.73	-10.5%	
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8596	-3.3%	
4.63	-5.6%	
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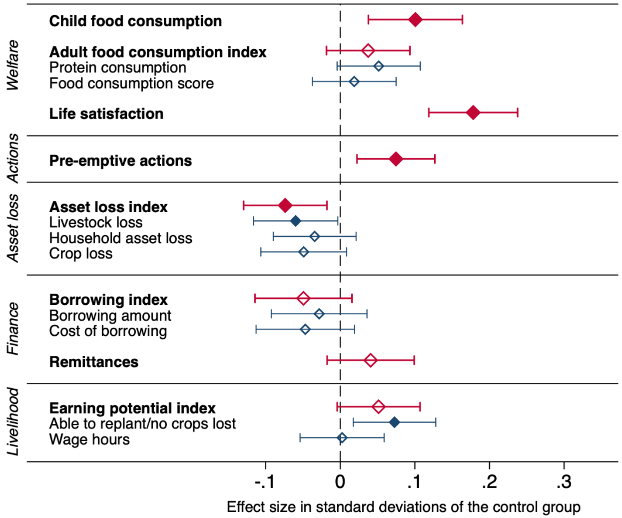

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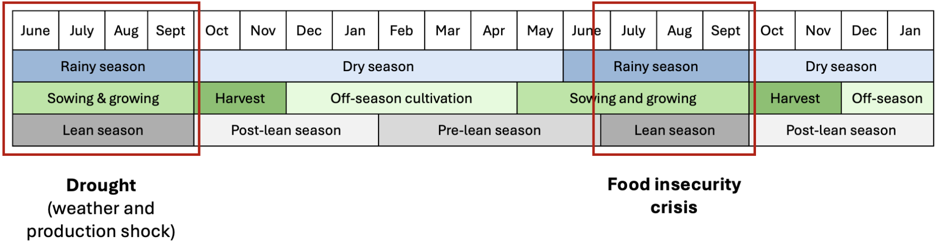
2. Early response to drought in Niger

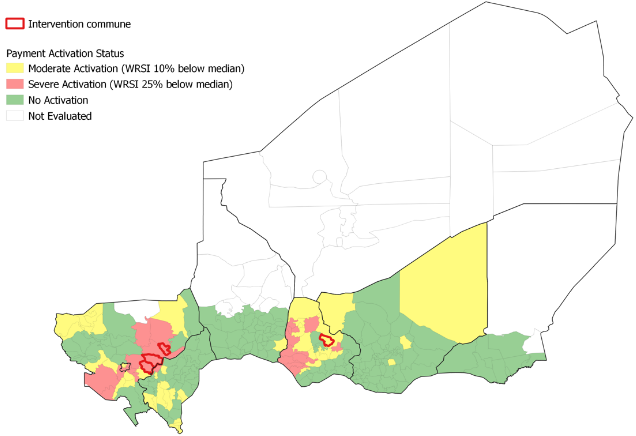
Early vs traditional response to slow-onset shocks

- Droughts amplify the effects of seasonality on income and consumption, especially for farmers dependent on rain-fed agriculture.



- The optimal timing of response to drought is ambiguous.
  - Food insecurity is expected to be highest during the lean season.
  - But it may be rational for households to increase consumption earlier if prices are lower, labor needs or returns to nutrition higher, and some smoothing is possible.

Shock-responsive transfer pilot and 2021 drought in Niger



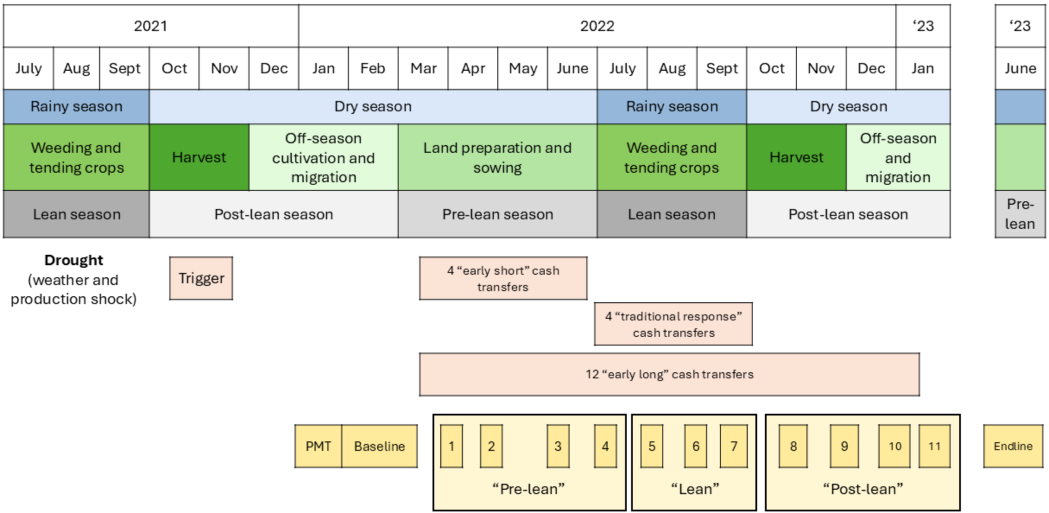
Dry spells in 2021 agricultural season caused cereal production to drop by 40% (FEWS NET)

Main trigger: Water Requirement Satisfaction Index (WRSI, October 2021)

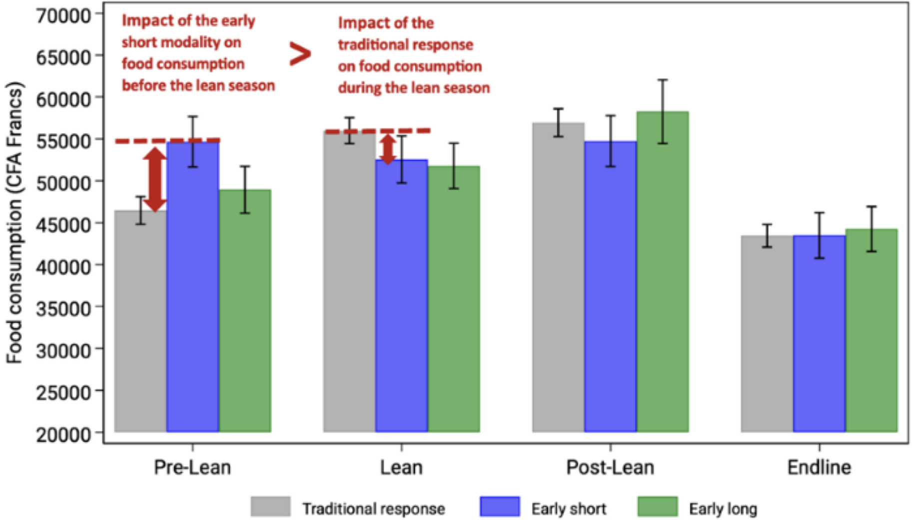
Coverage

- 4 communes activated; 3 in RCT
- 15,400 households targeted (poorest 44% based on PMT data)
- 171 villages and 4144 households in study sample.

Timing of interventions, high-frequency and endline surveys

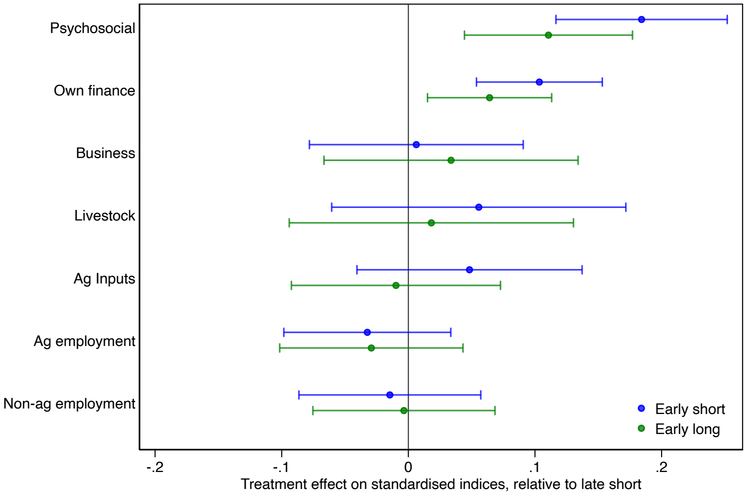


Impacts on food consumption in each period



Transfer timing has no effect on livelihoods

Pre-lean season



3. Policy takeaways

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3. The anticipatory cash transfer enabled households **to take different decisions** that altered the flood and drought impacts at a critical juncture in time.

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4. Varying the timing of cash transfers is **unlikely to build longer-term resilience**.

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    - Coordination mechanisms

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3. Pre-arranged financing needs to be readily available for the response. Governments should consider disaster funds, insurance or contingent credit.